SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: MLA03
COURSE CODE: Reinforcement learning for Environment and Agent

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COURSE OUTCOME COVERED

CO2: *Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Dr G.A. SENTHIL

QUESTIONS: 05 **Total Marks: 50**

Code of Conduct:

*I __**Bhanu Teja/192472259**_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Bhanu Teja

Annexure A

Application - Based Questions

1. Design and develop a Reinforcement Learning framework a ride-sharing company uses Reinforcement Learning to match drivers with passengers efficiently. Apply RL concepts to define the state space, action space, and reward function to minimize waiting time and improve service quality.

1. Objective of the Ride-Sharing System

The objective of the ride-sharing system is to efficiently match available drivers with passenger requests while minimizing passenger waiting time, reducing travel distance, and improving service quality. The system should also maximize driver utilization, reduce idle time, and increase customer satisfaction by making intelligent driver assignment decisions.

2. Why Reinforcement Learning is Suitable

Reinforcement Learning (RL) is suitable because the ride-sharing environment changes continuously due to traffic conditions, passenger requests, driver availability, and weather conditions. Unlike traditional rule-based systems, RL learns from experience and continuously improves its decisions. The RL agent receives feedback after every ride and updates its policy to make better driver assignments in future requests.

3. Agent and Environment

Agent

The RL agent is the Driver Matching System that decides which driver should be assigned to a passenger.

Environment

The environment includes:

- Available drivers
- Passenger requests
- GPS locations
- Traffic conditions
- Road network
- Ride completion status

The agent interacts with this environment to make optimal decisions.

4. State Space

The state space contains all information required by the RL agent before making a decision.

It includes:

- Driver location
- Passenger pickup location
- Passenger destination
- Driver availability
- Traffic condition
- Estimated travel time
- Driver rating
- Vehicle type

Example State

Parameter	Value
Driver Location	Anna Nagar
Passenger Location	T Nagar
Driver Available	Yes
Traffic	Medium
Estimated Arrival	4 Minutes

5. Action Space

The RL agent can perform the following actions:

- Assign nearest driver
- Assign another nearby driver
- Reject ride request
- Reassign driver
- Delay assignment

The selected action depends on the current system state.

6. Reward Function

The reward function encourages better driver-passenger matching.

Situation	Reward
Successful ride assignment	+100

Situation	Reward
Passenger waiting time < 5 min	+50
Short travel distance	+30
High customer rating	+20
Driver cancellation	-30
Long waiting time	-50
Wrong driver assignment	-40

The RL agent tries to maximize the total reward.

7. Policy Used by the RL Agent

The policy is the decision-making strategy followed by the RL agent.

Example policy:

- If the nearest driver is available and traffic is low, assign that driver.
- If traffic is high, assign another driver with a shorter estimated travel time.
- If no nearby driver is available, search within a larger service area.
- If a driver cancels, immediately reassign another nearby driver.

The policy is updated after every completed ride.

8. Learning Process

The RL agent learns through interaction with the environment.

Exploration

Initially, the agent tries different driver assignment strategies to understand which decisions give better rewards.

Exploitation

After learning, the agent uses the best-performing strategy more frequently.

After every completed ride, the agent receives feedback such as:

- Passenger waiting time
- Ride completion
- Customer rating
- Driver response
- Travel time

Using this feedback, the RL agent continuously updates its policy and improves future matching decisions.

9. RL Architecture Diagram

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   Passenger Request
        |
        ▼
 Ride-Sharing Environment
        |
        ▼
      RL Agent
        |
   Select Best Driver
        |
        ▼
   Driver Assignment
        |
        ▼
Ride Completed & Feedback
        |
        ▼
   Reward / Penalty
        |
        ▼
    Policy Updated
```

10. Expected Outcomes

Applying Reinforcement Learning in ride-sharing systems provides the following benefits:

- Reduced passenger waiting time.
- Faster driver allocation.
- Efficient driver utilization.
- Lower fuel consumption.
- Improved customer satisfaction.
- Reduced ride cancellations.

- Better route optimization.
- Continuous improvement through learning.

2. An e-learning platform uses Reinforcement Learning to recommend personalized study materials. Recall how exploration and exploitation strategies help balance familiar and new content recommendations.

1. Objective of the E-Learning System

The objective of the e-learning platform is to recommend personalized study materials that match each student's learning ability, interests, and progress. The Reinforcement Learning (RL) system aims to improve learning outcomes by providing the most relevant content while maintaining student engagement. It also helps students discover new learning materials without overwhelming them.

2. Why Reinforcement Learning is Suitable

Reinforcement Learning is suitable because every student has different learning preferences, strengths, and weaknesses. A student's interests and performance change over time, making fixed recommendation systems less effective.

The RL agent continuously observes student behavior, recommends study materials, receives feedback based on student interactions, and updates its recommendation strategy. This allows the platform to adapt to each student's learning pattern and provide more accurate recommendations over time.

3. Agent and Environment

Agent

The RL Agent is the recommendation system that selects the most appropriate study material for each student.

Environment

The environment includes:

- Student profile
- Learning history
- Quiz scores
- Course progress
- Learning preferences

- Available study materials
- Student feedback

The agent interacts with this environment to improve future recommendations.

4. State Space

The state space represents the current learning status of a student.

It includes:

- Student ID
- Current course
- Completed lessons
- Quiz performance
- Preferred learning style
- Time spent studying
- Difficulty level
- Learning progress

Example State

Parameter	Value
Student	John
Course	Python Programming
Quiz Score	82%
Preferred Content	Videos
Completed Lessons	12
Difficulty Level	Intermediate

The RL agent uses this information before recommending the next study material.

5. Action Space

The RL agent can perform the following actions:

- Recommend a familiar topic.
- Recommend a new topic.
- Suggest practice quizzes.
- Recommend video lectures.
- Recommend PDF notes.

- Recommend coding exercises.
- Suggest revision material.

The selected action depends on the student's current learning progress.

6. Reward Function

The reward function helps the RL agent recommend useful learning materials.

Student Response	Reward
Student completes lesson	+100
High quiz score	+50
Student watches entire video	+30
Student practices exercises	+20
Student skips recommendation	-20
Student leaves course	-50
Poor quiz performance	-30

The RL agent tries to maximize the overall learning success of the student.

7. Exploration and Exploitation Strategy

Exploration

Exploration means recommending new or different study materials that the student has not used before.

Examples:

- Suggesting a new programming language.
- Recommending advanced topics.
- Introducing interactive learning activities.
- Providing new learning resources.

Exploration helps students discover new knowledge and prevents repetitive learning.

Exploitation

Exploitation means recommending study materials that have previously helped the student learn effectively.

Examples:

- Recommending similar video lectures.
- Providing practice questions from familiar topics.

- Suggesting revision notes based on previous performance.

Exploitation improves learning efficiency by using successful recommendations.

Balancing Exploration and Exploitation

The RL system balances both strategies:

- Too much exploration may confuse students with unfamiliar content.
- Too much exploitation may limit learning opportunities by repeatedly recommending the same type of material.

A balanced approach ensures both effective learning and knowledge expansion.

8. Learning Process

The RL agent continuously learns from student interactions.

After each recommendation, it collects feedback such as:

- Lesson completion
- Quiz scores
- Time spent learning
- Number of repeated lessons
- Student ratings
- Course completion rate

Based on this feedback, the RL agent updates its recommendation policy and improves future suggestions.

9. RL Architecture Diagram

Student Login

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▼

E-Learning Environment

(Course, Progress, Quiz Scores)

|

▼

RL Recommendation Agent

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Recommend Study Material

|

▼

Student Studies / Gives Feedback

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▼

Reward / Penalty Generated

|

▼

Update Recommendation Policy

10. Expected Outcomes

Applying Reinforcement Learning in an e-learning platform provides the following benefits:

- Personalized learning experience.
- Better student engagement.
- Higher course completion rate.
- Improved academic performance.
- Balanced recommendation of familiar and new content.
- Continuous improvement in recommendation accuracy.
- Increased student satisfaction.
- Adaptive learning based on individual progress.

Conclusion

Reinforcement Learning enables e-learning platforms to provide personalized study material by continuously learning from student interactions. Through the balanced use of exploration and exploitation, the RL agent recommends both familiar and new learning resources, helping students strengthen existing knowledge while discovering new concepts. Over time, the system adapts to each student's learning style, resulting in improved engagement, higher learning efficiency, and better academic performance.

3. Design and develop a Reinforcement Learning framework a robotic vacuum cleaner uses Reinforcement Learning to clean efficiently while avoiding obstacles. Apply the concept of policy and explain how the value function improves performance over time.

1. Objective of the Robotic Vacuum Cleaner

The objective of the robotic vacuum cleaner is to clean the maximum area of a room while avoiding obstacles, minimizing battery consumption, and reducing cleaning time. The Reinforcement Learning (RL) agent learns the best cleaning path through interaction with the environment. The system aims to improve cleaning efficiency, avoid collisions, and maximize overall cleaning performance.

2. Why Reinforcement Learning is Suitable

Reinforcement Learning is suitable because the environment is dynamic and unpredictable. Furniture, walls, people, and other objects may change position, making fixed cleaning paths ineffective.

The RL agent continuously observes the surroundings, selects appropriate actions, receives rewards based on cleaning performance, and updates its policy. This enables the robot to adapt to different room layouts and improve its cleaning strategy over time.

3. Agent and Environment

Agent

The RL Agent is the robotic vacuum cleaner that decides where to move and how to clean efficiently.

Environment

The environment includes:

- Room layout
- Obstacles (Furniture, Walls)
- Dirty areas
- Charging station
- Battery level
- Cleaning status

The agent interacts with the environment to complete the cleaning task efficiently.

4. State Space

The state space represents the current condition of the robotic vacuum cleaner and its surroundings.

It includes:

- Current robot location
- Obstacle position
- Dirt location
- Battery level
- Cleaning progress
- Distance to charging station
- Remaining uncleaned area

Example State

Parameter	Value
Robot Position	Living Room
Dirt Detected	Yes
Obstacle Ahead	Chair
Battery Level	70%
Cleaned Area	65%

The RL agent observes this information before selecting the next action.

5. Action Space

The RL agent can perform the following actions:

- Move Forward
- Turn Left
- Turn Right
- Clean Current Area
- Avoid Obstacle
- Return to Charging Station
- Stop Cleaning

Each action changes the state of the environment.

6. Reward Function

The reward function encourages efficient cleaning and obstacle avoidance.

Situation	Reward
Clean dirty area	+100

Situation	Reward
Clean new area	+50
Successfully avoid obstacle	+30
Return to charging station safely	+20
Hit obstacle	-50
Re-clean same area	-20
Battery exhausted before charging	-100
Idle without cleaning	-10

The RL agent learns to maximize the total reward while minimizing unnecessary movements and collisions.

7. Policy and Value Function

Policy

A policy is the strategy used by the RL agent to decide the next action.

Example Policy:

- If dirt is detected nearby → Clean the area.
- If an obstacle is detected → Turn left or right.
- If battery level is low → Return to the charging station.
- If the room is clean → Stop cleaning.

The policy improves continuously after every cleaning cycle.

Value Function

The Value Function estimates how beneficial it is for the robot to be in a particular state.

Initially, the robot has no knowledge of the environment. After multiple cleaning sessions, it learns:

- Which paths clean more area.
- Which routes contain obstacles.
- Where dirt is commonly found.
- When to return for charging.

The value of good states increases over time, allowing the robot to choose more efficient cleaning paths and improve overall performance.

8. Learning Process

The RL agent learns through continuous interaction with the environment.

Exploration

Initially, the robot explores different areas and routes to understand the room layout and obstacle locations.

Exploitation

After learning, the robot follows the best cleaning routes and avoids unnecessary movements.

The RL agent continuously updates its knowledge using:

- Cleaning success
- Battery usage
- Obstacle avoidance
- Cleaning time
- Area coverage

This continuous learning helps improve cleaning efficiency over time.

9. RL Architecture Diagram

Dirty Room
|
▼
Cleaning Environment
(Dirt, Obstacles, Battery Level)
|
▼
Reinforcement Learning Agent
(Robotic Vacuum Cleaner)
|
Select Cleaning Action
|
▼
Move • Clean • Avoid Obstacle
|
▼
Reward / Penalty Received

│
▼
Update Policy & Value Function
│
└──────────────► Repeat

10. Expected Outcomes

Applying Reinforcement Learning to the robotic vacuum cleaner provides the following benefits:

- Efficient cleaning path.
- Better obstacle avoidance.
- Reduced cleaning time.
- Lower battery consumption.
- Maximum area coverage.
- Intelligent navigation.
- Continuous improvement through learning.
- Longer robot lifespan.
- Improved cleaning quality.
- Better user satisfaction.

Conclusion

Reinforcement Learning enables a robotic vacuum cleaner to clean efficiently by learning from its interactions with the environment. The RL agent observes the room conditions, selects suitable actions, receives rewards, and continuously updates its policy and value function. Over time, the robot identifies the most efficient cleaning paths, avoids obstacles, conserves battery power, and improves cleaning performance. This adaptive learning approach makes RL an effective solution for autonomous robotic cleaning systems.

4. A cryptocurrency trading system uses Reinforcement Learning to decide buy, sell, or hold actions. Outline the challenges in designing reward functions and explain how poor reward design can affect performance.

1. Objective of the Cryptocurrency Trading System

The objective of the cryptocurrency trading system is to maximize profit by making intelligent decisions on when to **buy**, **sell**, or **hold** cryptocurrencies. The Reinforcement Learning (RL) agent continuously analyzes market conditions and learns the best trading strategy while minimizing financial risks and losses. The system aims to improve trading accuracy, increase returns, and adapt to rapidly changing market conditions.

2. Why Reinforcement Learning is Suitable

Cryptocurrency prices change rapidly due to market demand, global events, investor behavior, and economic conditions. Traditional rule-based trading systems cannot easily adapt to these continuous market fluctuations.

Reinforcement Learning is suitable because it learns from market interactions. The RL agent observes market conditions, performs trading actions, receives rewards based on profit or loss, and updates its trading strategy. Over time, the agent learns to make better investment decisions and improves trading performance without relying on fixed rules.

3. Agent and Environment

Agent

The **RL Agent** is the cryptocurrency trading system that decides whether to **Buy**, **Sell**, or **Hold** a cryptocurrency.

Environment

The environment includes:

- Cryptocurrency market prices
- Trading volume
- Market trends
- Historical price data
- News and market sentiment
- Investor demand
- Trading portfolio

The RL agent interacts with this environment to maximize long-term profit.

4. State Space

The **state space** represents the current condition of the cryptocurrency market before making a trading decision.

It includes:

- Current cryptocurrency price
- Historical price trend
- Market volatility
- Trading volume
- Portfolio balance
- Available cash
- Number of coins owned
- Time of trading

Example State

Parameter	Value
Cryptocurrency	Bitcoin
Current Price	₹48,00,000
Market Trend	Upward
Trading Volume	High
Coins Owned	2 BTC
Portfolio Balance	₹12,00,000

The RL agent analyzes this information before selecting an action.

5. Action Space

The RL agent can perform the following actions:

- Buy cryptocurrency
- Sell cryptocurrency
- Hold the current investment
- Buy additional coins
- Sell a portion of holdings

Each action changes the portfolio and future trading opportunities.

6. Reward Function

The reward function encourages profitable trading decisions.

Situation	Reward
Profitable trade	+100
Correct Buy decision	+50
Correct Sell decision	+50
Long-term profit	+30
Small loss	-20
Large financial loss	-100
Unnecessary trading	-30
Holding during market crash	-50

The RL agent tries to maximize total profit while minimizing trading losses.

7. Policy Used by the RL Agent

The **policy** is the decision-making strategy used by the RL agent.

Example Policy

- If the market trend is rising, **Buy**.
- If the market trend is falling, **Sell**.
- If the market is uncertain, **Hold**.
- If profit reaches the target, **Sell** and secure gains.
- If the risk level is high, avoid unnecessary trading.

The policy is continuously updated after every trade.

8. Learning Process and Challenges in Reward Design

The RL agent learns from every trading decision.

Exploration

Initially, the agent explores different trading strategies by trying various buy, sell, and hold actions to understand market behavior.

Exploitation

After sufficient learning, the agent uses the trading strategy that has produced the highest rewards in previous experiences.

Challenges in Designing Reward Functions

Designing an effective reward function is one of the biggest challenges in Reinforcement Learning.

Some challenges include:

- Rewarding only short-term profits may ignore long-term investment growth.
- Poor reward design can encourage excessive buying and selling.
- Market fluctuations can produce misleading rewards.
- High-risk trades may appear profitable in the short term but lead to long-term losses.
- Multiple objectives such as profit, risk reduction, and transaction cost must be balanced.

Effects of Poor Reward Design

If the reward function is poorly designed:

- The agent may make risky investment decisions.
- Frequent unnecessary trading may increase transaction costs.
- The system may ignore market risks.
- Long-term profitability may decrease.
- The learned trading strategy may become unstable and unreliable.

Therefore, the reward function should encourage consistent profits while controlling financial risk.

9. RL Architecture Diagram

Cryptocurrency Market

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▼

Trading Environment

(Prices, Trends, Volume, Portfolio)

|

▼

Reinforcement Learning Agent

|

Buy / Sell / Hold Decision

|

▼

Trading Result (Profit/Loss)

|

▼

Reward or Penalty

│

▼

Update Trading Policy

│

└────────────► Repeat

10. Expected Outcomes

Applying Reinforcement Learning in cryptocurrency trading provides the following benefits:

- Intelligent trading decisions.
- Increased trading profits.
- Reduced financial risk.
- Better market prediction.
- Adaptive trading strategy.
- Faster response to market changes.
- Continuous improvement through learning.
- Better portfolio management.
- Improved investment efficiency.
- Higher long-term returns.

Conclusion

Reinforcement Learning provides an intelligent approach to cryptocurrency trading by learning from market interactions and continuously improving trading strategies. The RL agent analyzes market conditions, selects appropriate buy, sell, or hold actions, and updates its policy using rewards based on trading performance. A well-designed reward function is essential for achieving stable and profitable trading, while poor reward design can lead to risky decisions and reduced overall performance. Therefore, Reinforcement Learning offers an effective solution for adaptive and intelligent cryptocurrency trading systems.

5. Develop a smart grid system uses Reinforcement Learning to manage electricity distribution. Create an RL model by defining state space, action space, and reward mechanism.

1. Objective of the Smart Grid System

The objective of the smart grid system is to efficiently distribute electricity to consumers while maintaining a balance between electricity generation and demand. The Reinforcement Learning (RL) system aims to reduce energy wastage, prevent power overload, minimize power outages, and improve the reliability of electricity distribution. It also helps integrate renewable energy sources such as solar and wind power into the electrical grid.

2. Why Reinforcement Learning is Suitable

Traditional electricity distribution systems use predefined rules and schedules to allocate power. These methods cannot quickly respond to sudden changes in electricity demand, equipment failures, or renewable energy fluctuations.

Reinforcement Learning is suitable because it continuously learns from real-time data such as electricity demand, weather conditions, energy production, and grid status. The RL agent observes the current state of the grid, selects the best distribution strategy, receives rewards based on system performance, and updates its policy. This allows the smart grid to make intelligent decisions and improve electricity management over time.

3. Agent and Environment

Agent

The **RL Agent** is the Smart Grid Controller responsible for deciding how electricity should be distributed across different regions.

Environment

The environment consists of:

- Power generation stations
- Transmission lines
- Consumers (Residential, Commercial, Industrial)
- Renewable energy sources
- Electricity demand
- Weather conditions
- Battery storage systems

The RL agent interacts with this environment to ensure efficient electricity distribution.

4. State Space

The **state space** represents the current condition of the smart grid before making a decision. It includes:

- Current electricity demand
- Available electricity supply
- Battery storage level
- Solar and wind energy generation
- Time of day
- Weather conditions
- Grid load
- Power outage status

Example State

Parameter	**Value**
Electricity Demand	High
Available Supply	120 MW
Solar Generation	40 MW
Battery Storage	75%
Grid Load	Medium
Weather	Sunny

The RL agent analyzes this information before deciding how to distribute electricity.

5. Action Space

The RL agent can perform the following actions:

- Increase electricity supply.
- Reduce electricity supply.
- Store excess electricity in batteries.
- Distribute renewable energy.
- Switch electricity between grid zones.
- Activate backup power sources.
- Balance load among substations.

Each action affects the efficiency and stability of the power grid.

6. Reward Function

The reward function encourages efficient electricity distribution while minimizing energy loss and outages.

Situation	Reward
Balanced electricity supply	+100
Reduced energy wastage	+50
Efficient use of renewable energy	+40
Stable grid operation	+30
Power outage	-100
Grid overload	-80
Energy wastage	-40
Uneven power distribution	-30

The RL agent learns to maximize the total reward by maintaining a stable and efficient electricity distribution system.

7. Policy Used by the RL Agent

A **policy** is the strategy used by the RL agent to decide how electricity should be distributed.

Example Policy

- If electricity demand is high, increase power supply from available sources.
- If renewable energy generation is high, prioritize solar and wind power.
- If battery storage is full, distribute excess energy to the grid.
- If grid load becomes excessive, redistribute power to nearby substations.
- If demand decreases, store excess electricity for future use.

The policy is continuously updated based on system performance and changing demand.

8. Learning Process

The RL agent learns through continuous interaction with the smart grid.

Exploration

Initially, the RL agent explores different electricity distribution strategies by allocating power in different ways and observing the outcomes.

Exploitation

After learning from previous experiences, the RL agent uses the most effective distribution strategy to maintain grid stability and maximize efficiency.

The RL agent continuously improves its decisions by learning from:

- Electricity demand patterns
- Renewable energy production
- Battery usage
- Grid performance
- Power outage history
- Energy consumption trends

This continuous learning enables the smart grid to adapt to changing conditions and improve electricity management over time.

9. RL Architecture Diagram

Electricity Demand

|

▼

Smart Grid Environment

(Consumers, Renewable Energy, Grid Load,

Weather, Battery Storage, Power Plants)

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▼

Reinforcement Learning Agent

(Grid Controller)

|

Select Distribution Strategy

|

▼

Supply Power / Store Energy / Balance Load

|

▼

Reward or Penalty Received

|

▼

Update RL Policy

Repeat

10. Expected Outcomes

Applying Reinforcement Learning to a smart grid system provides several benefits:

- Efficient electricity distribution.
- Reduced power outages.
- Better load balancing.
- Lower energy wastage.
- Improved use of renewable energy.
- Intelligent demand forecasting.
- Better battery management.
- Reduced operational costs.
- Continuous improvement through learning.
- Increased reliability of the power grid.

Conclusion

Reinforcement Learning provides an intelligent and adaptive solution for smart grid management. The RL agent continuously monitors electricity demand, available power, renewable energy generation, and grid conditions to make optimal distribution decisions. By defining appropriate **state spaces**, **action spaces**, and **reward mechanisms**, the smart grid can efficiently balance electricity supply and demand while minimizing energy losses and power interruptions. Continuous learning allows the system to adapt to changing consumption patterns and improve the overall reliability and efficiency of electricity distribution.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand